Reg No.:_______________ Name:__________________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: EET303

Course Name: MICROPROCESSORS AND MICROCONTROLLERS

Max. Marks: 100 Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks) Marks

1	List the control and status signals of 8085.	(3)
2	Explain the operation of the following 8085 instructions: (i) DAD D (ii) INR M (iii) XCHG.	(3)
3	Define PUSH and POP instructions of 8085.	(3)
4	Calculate the loop delay if the register pair BC of 8085 microprocessor is loaded with 8000H. Assume the system clock frequency to be 3.072 MHz.	(3)
5	Explain maskable and non-maskable interrupts of 8085.	(3)
6	Distinguish between hard and soft real-time systems.	(3)
7	Explain the operation of the following 8051 instructions: (i) MOV A,@R_1 (ii) CPL A (iii) SWAP A.	(3)
8	Define the assembler directives of 8051.	(3)
9	List the interrupts of 8051.	(3)
10	Explain the TMOD register of 8051.	(3)

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

11	a)	Explain the architecture of 8085 microprocessor with the help of a neat functional block diagram.	(10)
	b)	Identify the machine cycles of the following instructions: (i) MOV B, M (ii) MVI A, 05H (iii) STA 2065H (iv) OUT 01H.	(4)
12	a)	Sketch and explain the timing diagram of the instruction MOV M, A with opcode 77H stored in the memory location 4500H.	(9)
	b)	Explain the addressing modes of 8085 with suitable examples.	(5)

Module -2

13 a) Ten bytes of data are stored in consecutive memory locations starting at 2050H. With suitable explanations, write an assembly language program in 8085 to transfer the entire block of data to new memory locations starting at 2070H. (9)

b) The contents of accumulator and C register are 87H and 79H respectively. Find the contents of the accumulator and flags after the execution of the instruction ADD C. (5)

14 a) Write and explain an assembly language program in 8085 to perform BCD to Binary conversion. (9)

b) Explain the significance of stack memory while executing CALL and RET instructions in 8085. (5)

Module -3

15 a) Explain the interfacing of seven segment LED with 8085. (8)

b) Draw the control word format for the I/O mode of 8255. Hence frame the control word for the following configuration in Mode 0 operation: (6)
Ports A, B and C_L - output ports, Port C_U- input port.

16 a) With the help of a neat sketch, explain the register organization and SFR of 8051. (10)

b) Differentiate between microprocessors and microcontrollers. (4)

Module -4

17 a) Classify 8051 instructions into various categories. Give examples for each. (8)

b) Explain the addressing modes of 8051 with suitable examples. (6)

18 a) Define data types in C and explain (6)

b) Write an 8051 C program to toggle all the bits of P0 and P2 continuously with a 250ms delay. (8)

Module -5

19 a) Explain SCON and SBUF registers in 8051. (7)

b) Explain Interrupt Enable and Interrupt Priority registers of 8051. (7)

20 a) Design a DAC interfacing circuit with 8051. (8)

b) Write an 8051 program to transfer letter “A” serially at 4800 baud rate. (6)

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